

Kenya Market Pricing & Rate Card

A ground-up pricing model for the Cirqen Labs biomedical engineering platform — ISO/IEC 17025-aligned calibration, equipment inventory, planned maintenance, and offline-first multi-site synchronization — benchmarked against Kenyan enterprise-software norms and international CMMS/EAM pricing.

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Currency: Kenyan Shilling (KES), reference rate 1 USD $\approx$ 129.4 KES

Basis: Direct source-code audit ($\approx$ 110,800 lines, 16 application modules) + Kenyan & international market comparables

Scope: Recommended pricing structure, tiered rate card, add-ons, sample cost scenarios

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1. Executive Summary

Cirqen Labs is a Django-based biomedical engineering platform combining **equipment inventory management**, an **ISO/IEC 17025-aligned calibration and certification engine**, **planned preventive maintenance (PPM) scheduling**, **job-card/workshop management**, and an **offline-first multi-site synchronization layer** that lets satellite facilities operate without connectivity and reconcile with a central headquarters (HQ) server. This is materially more specialized than a generic CMMS: the calibration engine performs decimal-precision uncertainty-budget mathematics and mints gapless, non-reusable certificate numbers under a single serialized authority — functionality no off-the-shelf Kenyan ERP or generic maintenance tool offers.

~110,800 LINES OF PYTHON ACROSS 16 APPS	KES 480K–2.88M RECOMMENDED ANNUAL SUBSCRIPTION RANGE	3 tiers FACILITY → REFERRAL → ENTERPRISE NETWORK
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The recommendation is a **tiered annual-subscription model with a one-time implementation fee** — the dominant pattern in both the Kenyan SaaS/ERP market and international CMMS/EAM pricing — priced *per facility and per equipment population* rather than strictly per user, because Kenyan hospital procurement budgets are held at the facility/departmental level, not the individual-seat level. Three tiers are proposed: **Facility Starter** (single site, core inventory/PPM), **Referral Professional** (adds the full calibration & certification engine), and **Enterprise Network** (adds offline-first multi-site sync with a central certificate authority — the product's most differentiated and defensible capability). A perpetual-license path is also modeled for public-sector tenders that require capex procurement rather than recurring opex.

2. What the System Actually Does

Findings below come from reading the codebase directly, not from marketing copy.

Module	Function	Why it matters commercially
CalSoft (12,644 LOC)	Full calibration workflow: procedure definitions, tolerance & uncertainty-budget mathematics in <code>decimal.Decimal</code> (not floating point), pass/fail rules, independent-reviewer approval, gapless certificate numbering, certificate PDF rendering.	This is the feature that lets a hospital stop outsourcing calibration certification to third-party labs for in-house-serviceable equipment — the single strongest revenue justification in the product.
calSchedules (8,807 LOC)	Calendar-based, group-aware PPM/calibration scheduling; a machine calibrated early/late stays credited to its planned month.	Drives the compliance dashboards auditors (ISO 17025 assessors, county health inspectors) actually check.
sync (17,988 LOC)	Offline-first sync agent: outbox pattern, LWW conflict resolution with total ordering, schema-drift detection, conflict quarantine, backoff-aware upload throttling.	Directly addresses Kenya's connectivity reality — rural/county facilities can keep working offline and reconcile later. This is the hardest engineering in the repo and the clearest enterprise-tier differentiator.
hq_server	Central certificate-numbering authority via PostgreSQL advisory locks; single source of truth across sites.	Required infrastructure for any multi-site deployment — packaged as part of the Enterprise tier.
Inventory (8,445 LOC)	Equipment register: department, manufacturer, equipment description/asset records.	Baseline data layer every tier needs.
ppms / jobcard / workshop	Preventive-maintenance scheduling, work orders with spare-parts tracking, workshop records.	Core CMMS functionality — comparable to what international tools charge \$45–120/user/month for.
machineReports / reporthub	Equipment status reports, repair history, drift analysis, general reporting.	Management/compliance reporting layer.
parts_tools	Tools & accessories inventory with request/approval workflow.	Workshop logistics — natural upsell for larger facilities.
users / accounts / audit_log	Custom user model, e-signatures, security logs, append-only audit trail, sync-client tracking.	Table-stakes for any auditable regulated-industry software.

Positioning takeaway: the product sits at the intersection of two categories international buyers price separately — general CMMS/EAM (\$45–200/user/month) and dedicated calibration-management software (typically sold as a premium module or a standalone enterprise product). Bundling both, plus purpose-built offline sync for low-connectivity sites, is the core pricing argument for this market.

3. Kenyan Market Context

3.1 Structural buyer landscape

Kenya has 47 counties, each operating at least one county referral hospital plus a network of sub-county facilities, on top of national referral/teaching hospitals (KNH, MTRH) and a private tier (Aga Khan University Hospital, Nairobi Hospital, Mater, etc.). Biomedical engineering departments are chronically understaffed relative to equipment population: **Kenyatta National Hospital manages roughly 1,200 distinct types of sophisticated medical equipment with only about 24 biomedical engineers**. That ratio — not a lack of demand — is the real ceiling on what manual, paper-based calibration and maintenance tracking can achieve, and it is the strongest qualitative case for automation-driven pricing (the software substitutes for headcount the sector cannot hire fast enough).

3.2 What Kenyan buyers already pay for enterprise software

Reference point	Typical price (KES)	Source pattern
Fixed-price local ERP package (SME tier, e.g. ITKenya-style)	260,000 – 910,000	One-time / annual bundled package
Cloud SaaS ERP entry point	~6,500 / month	≈\$50/month equivalent
Mid-scale custom ERP/implementation project	1,000,000 – 5,000,000	One-time implementation
Large enterprise ERP implementation	5,000,000 – 50,000,000+	Multi-module, multi-site rollouts

This establishes the price ceiling and floor Kenyan procurement officers are already calibrated to. A single-site software product priced far outside the KES 260K–5M implementation band, or far outside low-hundred-thousands-per-year subscription band, will read as mispriced regardless of technical merit.

4. Competitive Benchmark (Converted to KES)

International CMMS pricing, converted at 1 USD = 129.4 KES, gives the ceiling this product should price under — since it must also absorb Kenya's lower ability-to-pay relative to US/EU healthcare IT budgets.

Product / tier	USD	KES equivalent	Notes
UpKeep — entry tier	\$20–45/user/mo	2,590–5,825/user/mo	Generic CMMS, no calibration engine
Fiix — Basic / Professional	\$45 / \$75/user/mo	5,825 / 9,705/user/mo	No offline-first multi-site sync
eMaint — Team / Professional / Enterprise	\$69 / \$85 / \$120/user/mo	8,929 / 11,000 / 15,528/user/mo	Enterprise tier still generic maintenance, not metrology-grade calibration
Mid-market team (10–30 users), blended	\$500–2,000/mo	64,700–259,000/mo 776,400–3,108,000/yr	Typical all-in monthly spend cited across CMMS vendors for a small-to-mid maintenance team
Dedicated calibration-management software	Sold separately, enterprise-quoted	—	International vendors treat ISO 17025 calibration workflow as a distinct, premium product line, not a CMMS feature

International tools charge *per user*, which punishes exactly the departments Kenya is trying to help — those with too few biomedical engineers per equipment unit. A per-facility model, priced below the international per-seat total for a comparable department, is both fairer and easier to sell into a 24-engineer, 1,200-device department like KNH's.

5. Pricing Philosophy

- **Per facility, not per seat.** Kenyan hospital IT/biomed budgets are approved at facility or departmental level. Uncapping user seats (within a fair-use ceiling) removes the friction of "will this block me from adding a technician" and matches local procurement psychology.
- **Tier by capability, not just by volume.** The calibration/certification engine (CalSoft) and the multi-site offline sync layer are the two components with no free local substitute — they are the paid upgrade path, not equipment-count thresholds alone.
- **One-time implementation fee + annual subscription**, mirroring the dominant Kenyan ERP pattern (fixed package + recurring) rather than a pure US-style monthly SaaS metered per user.
- **Price under the international per-seat total** for an equivalent department size, while pricing at or slightly above the generic local ERP band — justified by the ISO 17025 calibration engine and offline sync, which local ERPs do not offer at any price.
- **Offer both opex (subscription) and capex (perpetual license) paths**, because public-sector tenders in Kenya frequently require capital procurement rather than recurring vendor payments.

6. Rate Card — Subscription Tiers

Tier 1 — Facility Starter

KES 450,000 setup + KES 480,000/yr

Single site · up to 500 registered equipment items · up to 10 user seats · ≈ KES 40,000/month

Includes:

- Equipment inventory register (Inventory module)
- PPM & calibration-due scheduling (calSchedules, calendar/group-aware)
- Job cards, workshop & spare-parts tracking
- Basic reporting (Report Hub, Machine Reports)
- User accounts, e-signatures, audit log
- Standard business-hours support

Excludes: full ISO/IEC 17025 calibration & certificate engine, multi-site sync (available as add-ons, Section 7).

Target buyer: single sub-county or private hospital with a small biomedical engineering unit that needs to get off spreadsheets/paper.

Tier 2 — Referral Professional

KES 1,100,000 setup + KES 1,200,000/yr

Single site (or two co-located sites) · up to 2,500 equipment items · up to 30 user seats · ≈ KES 100,000/month

Everything in Tier 1, plus:

- Full CalSoft calibration & certification engine — procedure definitions, decimal-precision tolerance/uncertainty mathematics, independent-reviewer approval workflow, certificate PDF generation
- Single-site HQ server instance acting as the certificate-numbering authority
- Parts & tools inventory with request/approval workflow
- Drift analysis & extended machine reporting

Target buyer: county/national referral hospital or accredited private hospital biomedical engineering department that currently outsources calibration certification and wants to bring it in-house under its own ISO 17025 scope.

Tier 3 — Enterprise Network

KES 3,200,000 setup + KES 2,880,000/yr

Up to 5 sites included · unlimited equipment · unlimited seats (fair use) · +KES 420,000/site/yr beyond 5

Everything in Tier 2, plus:

- Offline-first multi-site sync agent: outbox pattern, convergent conflict resolution, schema-drift guard, conflict quarantine — satellite facilities keep working without connectivity and reconcile automatically on reconnect
- Central HQ certificate authority serving all sites (gapless, non-colliding certificate numbers network-wide)
- Cross-site dashboards and county/network-level compliance reporting
- Priority onboarding for each additional site
- Priority SLA (see Section 10)

Target buyer: a county health department rolling the platform out across its referral hospital plus its network of sub-county and dispensary-level facilities, or a private hospital group with multiple branches.

7. Add-On Modules & Services

Add-on	One-time	Recurring	Notes
Additional user seats beyond tier allowance	—	KES 1,800 / seat / mo	Tier 3 seats are fair-use uncapped
CalSoft calibration engine, added to Tier 1	KES 350,000	KES 25,000 / mo	Upgrade path without moving full tier
Multi-site sync, added to Tier 1 or 2	KES 550,000	KES 35,000 / mo / site	Includes HQ server provisioning
Legacy data migration (paper/Excel records)	KES 150,000 – 600,000	—	Banded by record volume
On-site training (2-day session, per site)	KES 95,000	—	Technologist + reviewer/approver tracks
Custom module / report development	KES 6,500 / hr	—	20-hour minimum engagement
Priority Support SLA upgrade	—	+25% of annual subscription	4-hour response, 24/7 (see Section 10)

8. Sample Customer Cost Scenarios (3-Year TCO)

Scenario A — County Referral Hospital (single site, ~800 equipment items)

Year	Setup	Subscription	Total
Year 1	1,100,000	1,200,000	2,300,000
Year 2	—	1,200,000	1,200,000
Year 3	—	1,200,000	1,200,000
3-year TCO		KES 4,700,000 (≈ \$36,300)	

Tier 2. Roughly in line with a "smaller / specific module" Kenyan ERP implementation, but includes a capability (in-house calibration certification) no local ERP offers at any price.

Scenario B — National Teaching/Referral Hospital (KNH-scale, ~1,200 equipment types, single campus + satellite clinics, 5 sites)

Year	Setup	Subscription	Total
Year 1	3,200,000	2,880,000	6,080,000
Year 2	—	2,880,000	2,880,000
Year 3	—	2,880,000	2,880,000
3-year TCO		KES 11,840,000 (≈ \$91,500)	

Tier 3 base (5 sites included). At an unstaffed ratio of ~1 biomedical engineer per 50 equipment types, this is the profile the offline-sync + calibration-authority combination is built for.

Scenario C — County Health Network (1 referral hospital HQ + 8 sub-county facilities = 9 sites)

Year	Setup	Subscription	Total
Year 1	3,200,000	2,880,000 + 1,680,000	7,760,000
Year 2	—	4,560,000	4,560,000
Year 3	—	4,560,000	4,560,000

3-year TCO

KES 16,880,000 (≈ \$130,400)

Tier 3 + 4 extra sites (420,000/site/yr). Positions comfortably inside the KES 5M–50M band Kenyan buyers already associate with large, multi-site enterprise software rollouts.

9. Perpetual-License Alternative (Procurement/Tender Path)

Public-sector tenders in Kenya (county health departments, national hospitals under public procurement rules) often require a one-time capital purchase rather than an ongoing vendor subscription. For these buyers, offer:

Tier	Perpetual license fee	Annual maintenance (18%)
Facility Starter	KES 1,440,000	KES 259,200 / yr
Referral Professional	KES 3,600,000	KES 648,000 / yr
Enterprise Network (5 sites)	KES 8,640,000	KES 1,555,200 / yr

License fee modeled at 3× the equivalent annual subscription (a standard SaaS-to-perpetual conversion multiple); 18%/year maintenance covers updates, bug fixes, and standard support — the conventional enterprise-software maintenance rate.

10. Support, SLAs & Payment Terms

10.1 Support tiers

Tier	Response time	Channel	Included in
Standard	2 business days	Email / phone, business hours	All tiers, base price
Priority	4 hours, 24/7	Email / phone / on-call	+25% of annual subscription; included by default in Tier 3

10.2 Payment terms

- Implementation/setup fee: 50% on signing, 50% on go-live.
- Annual subscription: billed annually in advance, or quarterly at a 5% premium for cash-flow-constrained public facilities.
- Additional-site fees (Tier 3): billed at site go-live, pro-rated to the master renewal date.

10.3 Indicative implementation timeline

Phase	Tier 1	Tier 2	Tier 3 (per site, after HQ)
Setup & configuration	2–3 weeks	4–6 weeks	1–2 weeks
Data migration	1–2 weeks	2–4 weeks	1–2 weeks
Training & go-live	1 week	1–2 weeks	3–5 days

11. Methodology, Sources & Assumptions

This document is a modeled pricing recommendation, not a verified or published Cirqen Labs price list. It was produced by (1) directly auditing the codebase in `/home/cirqen/Desktop/cirqen-labs` — 16 Django apps, ~110,800 lines of Python, reading model definitions and the calibration-module technical documentation to establish real feature scope; and (2) triangulating pricing against public market data gathered via web search on 2026-08-07:

- USD/KES reference rate ≈129.4 (exchange-rates.org, 6 Aug 2026)
- International CMMS pricing: Fiix, UpKeep, eMaint published tiers (\$20–120/user/month); blended mid-market spend of \$500–2,000/month for 10–30 user teams (ITQlick, Fabrico, Facilio, Limble pricing guides, 2026)
- Kenyan ERP/SaaS pricing: fixed local packages (KES 260,000–910,000), cloud entry pricing (~\$50/month), and large custom-implementation bands (KES 1M–50M+) (ITKenya, Nuva, Aqiq Solutions, 2026 guides)
- Kenyan biomedical engineering staffing context: KNH manages ~1,200 equipment types with ~24 biomedical engineers (Daily Nation / ICMHS reporting); 47 counties each operate at least one referral hospital

No figure in this document should be treated as a live vendor quote, a confirmed competitor price, or a guarantee of what any specific Kenyan institution would pay — actual willingness-to-pay should be validated with 3–5 target buyers (a mix of one private hospital, one county referral hospital, and one national referral hospital) before this rate card is finalized or published externally. Tier boundaries (equipment counts, seat counts, site counts) are starting points for negotiation, not hard technical limits in the software.